Progressive Mid-Stage RGB-Thermal Fusion for Lightweight Drone-Based Person Detection

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Abstract

Drone-mounted person detection for search-and-rescue demands both speed and robustness under degraded visibility. We tackle this by fusing RGB and thermal infrared streams inside a dual-backbone YOLOv8-nano detector. Three fusion points are compared on the RGBTDronePerson benchmark: an early stage that trains one network on interleaved modality samples, a mid-stage that merges intermediate feature maps through concatenation, and a late stage that ensembles two independent detectors with Weighted Boxes Fusion. Mid-stage wins on recall and F1, so we refine it with a three-phase freeze-unfreeze curriculum that first locks the pretrained backbones, partially opens upper layers, then fully fine-tunes all parameters. This progressive schedule lifts mAP@0.5 from 54.26% (mid-stage baseline) to 59.04% without changing the network topology. Quality-aware fusion (QFDet) and its progressive variant (ProgQFDet) are also tested; both fall short of the simpler concatenation approach, likely because halving the neck channel width to 64/128/256 removes capacity needed for the dataset's tiny targets. Aligning both backbones on LLVIP pretraining (Stream 2) outperforms mixing SeaDronesSee with LLVIP (Stream 1) for every mid-stage method. After fine-tuning on custom frames from a DJI Mavic 3 (District 9, Ho Chi Minh City) and NTUT AIoT Lab drone imagery, the best model reaches 85.07% mAP@0.5 with pseudo-thermal input generated by the OpenCV HOT colormap. The exported ONNX model runs at 45 FPS on a laptop CPU. A companion demo streams live video, overlays detections, and projects GPS coordinates onto a Leaflet satellite map.

Keywords: RGB-thermal fusion, person detection, YOLOv8, drone, search and rescue, progressive training

1 Introduction

Consumer quadcopters equipped with cameras have become standard tools in search-and-rescue (SAR) operations [1, 2]. A drone can survey hectares of terrain in minutes, but the bottleneck shifts to the operator: watching a continuous video feed is fatiguing, and small or camouflaged persons are easy to miss. Automated detection alleviates this bottleneck, yet conventional RGB detectors struggle at night, in smoke, or when clothing blends with the background.

Thermal infrared cameras sense emitted heat rather than reflected light, making human bodies visible regardless of illumination [3]. Mounting both an RGB and a thermal sensor on the same platform creates an opportunity to fuse the two data streams for more reliable detection. The open question is *where* in the processing pipeline to combine them and *how* to train the combined model effectively.

We address both questions with a systematic study on the RGBTDronePerson dataset [4]. Our contributions are:

1. A controlled comparison of early, mid, and late stage fusion built on the same YOLOv8-nano backbone, showing that mid-stage concatenation achieves the best precision-recall balance.
2. A three-phase progressive freeze-unfreeze training protocol that adds 4–5 points of mAP@0.5 over joint training, without any architectural change.
3. Evidence that aligning both backbone pretraining datasets (LLVIP for both RGB and thermal) helps the fusion module more than domain-matched pretraining (SeaDronesSee for RGB).
4. A domain-adaptation study on custom drone footage with pseudo-thermal generation via the HOT colormap, boosting mAP@0.5 from 59.04% to 85.07%.
5. An end-to-end ONNX-based demo with GPS geo-localization at 45 FPS on CPU.

The rest of this paper is organized as follows. Section 2 reviews related work on object detection, multispectral fusion, and drone-based SAR. Section 3 describes our architecture, fusion strategies, and progressive training protocol. Section 4 presents the datasets and exploratory data analysis. Section 5 reports experimental results. Section 6 analyzes why each strategy performs the way it does. Section 7 describes the demonstration system. Section 8 discusses limitations, and Section 9 concludes.

2 Related Work

2.1 Real-Time Object Detection

The YOLO family has driven real-time detection forward since Redmon et al.'s single-shot formulation [5]. The core idea of predicting bounding boxes and class probabilities in a single forward pass eliminated the two-stage bottleneck of region proposal networks like Faster R-CNN. Subsequent iterations introduced batch normalization, anchor boxes, multi-scale prediction, and a suite of training tricks (mosaic augmentation, label smoothing, cosine annealing) that collectively pushed detection accuracy on COCO [7] well beyond the original YOLO's capabilities.

YOLOv8 [6] represents the latest major release in the Ultralytics lineage. It introduces an anchor-free detection head with decoupled classification and regression branches, a C2f feature aggregation block that replaces the CSP bottleneck from earlier versions, and distribution focal loss for tighter box coordinate regression. The Feature Pyramid Network (FPN) [8] combined with a Path Aggregation Network (PAN) [9] forms the neck, enabling multi-scale feature fusion across three output strides (8, 16, 32).

Newer iterations such as YOLOv9, YOLOv10, and YOLO11 push accuracy further through architectural refinements, but at the time our experiments began YOLOv8 offered the most mature and best-documented codebase through the Ultralytics library. We select the nano variant (3.2M parameters per backbone) because SAR deployment targets edge hardware where inference cost matters more than marginal accuracy gains from larger backbones.

2.2 Multispectral Fusion for Pedestrian Detection

Fusion strategies for combining RGB and thermal modalities sit on a spectrum defined by where in the pipeline the two streams interact.

Early fusion approaches stack or interleave modalities before any feature extraction takes place. The simplest form concatenates the two images channel-wise to produce a 6-channel input (3 RGB + 3 thermal) fed into a single backbone. While conceptually straightforward, this forces the first convolutional layer to learn cross-modal correlations from raw pixel values, which is a difficult task given how different the two intensity distributions are.

Mid-level fusion methods inject cross-modal information into intermediate feature maps. Guan et al. [10] demonstrated that concatenation-based mid-fusion can outperform more elaborate operators when the individual backbones provide discriminative features. Li et al. [11] proposed illumination-aware weighting that adapts the fusion balance based on ambient lighting conditions. The QFDet paper [4] introduced quality-aware scoring, where a learned quality map per modality per spatial location controls how much each stream contributes to the fused representation. Our work adapts the QFDet concept to a YOLOv8-nano backbone and compares it against simpler alternatives.

Late fusion methods run separate detectors and merge predictions at the box level. Weighted Boxes Fusion (WBF) [12] clusters overlapping predictions by IoU and produces a weighted average box, which is gentler than hard non-maximum suppression. The advantage is that no custom model code is needed; the disadvantage is that the two detectors share no information during training and can only reconcile disagreements at the post-processing stage.

Most published comparisons evaluate one strategy in isolation. Our work tests all three on identical backbones, data, and evaluation protocol, enabling direct conclusions about which fusion point works best for drone-based person detection.

2.3 Drone-Based Person Detection and SAR

The RGBTDronePerson benchmark [4] pairs aligned RGB-thermal frames captured from a DJI Matrice 300 RTK at altitudes of 20–80 m. The dataset contains 6,125 image pairs with over 60,000 person and rider annotations. Published baselines reach up to 46.72% mAP@0.5 using ResNet50-FPN backbones with roughly 60M parameters. Maritime datasets like SeaDronesSee [13, 14] provide complementary aerial training data with swimmer annotations from open-water scenarios.

The gap between published results and the operational needs of SAR teams is substantial. A production detector must run in real time on hardware that fits in a ground-station laptop, handle degraded thermal input when genuine infrared cameras are unavailable, and provide geographic coordinates rather than pixel-space bounding boxes. Our system addresses all three requirements.

3 Methodology

3.1 Architecture Overview

All mid-stage configurations share the skeleton shown in Fig. 1: two YOLOv8-nano backbones extract features at three pyramid levels (P3 with 64 channels at stride 8, P4 with 128 channels at stride 16, P5 with 256 channels at stride 32), a fusion module merges them, an FPN+PAN neck refines multi-scale representations, and a single-class Detect head outputs bounding boxes and confidence scores. The backbone and neck follow the Ultralytics YOLOv8 nano specification. We chose the nano variant because the detection target is a single class and the model must ultimately run on edge hardware for real-time SAR applications.

3.2 Fusion Strategies

Early stage. No dedicated fusion module is used. RGB and thermal frames from each scene enter the training set as separate samples sharing identical ground-truth annotations. A single YOLOv8-nano processes each image independently during training; whatever cross-modal generalization occurs is implicit in shared weights. At inference, the model receives either an RGB or a thermal frame and produces detections from that single input. This serves as a lower-bound reference against which the dual-backbone methods are measured.

Mid-stage baseline. At each pyramid level k, the RGB and thermal feature maps are concatenated along the channel axis and projected by a 1×1 Conv-BN-SiLU block:

$$F_k^{\text{fused}} = \text{SiLU}\big(\text{BN}\big(\text{Conv}_{1\times1}([F_k^{\text{RGB}} \| F_k^{\text{Thr}}])\big)\big) \tag{1}$$

The resulting channels (128, 256, 512 at P3–P5) feed into the standard FPN+PAN neck. This design is deliberately minimal: no attention mechanism, no quality gating, no spatial realignment. If this simple approach already performs well, more complex modules add overhead without proportional benefit.

Late stage. Two independent YOLOv8-nano detectors (one RGB, one thermal) are trained separately using the Ultralytics training API. At inference their box sets

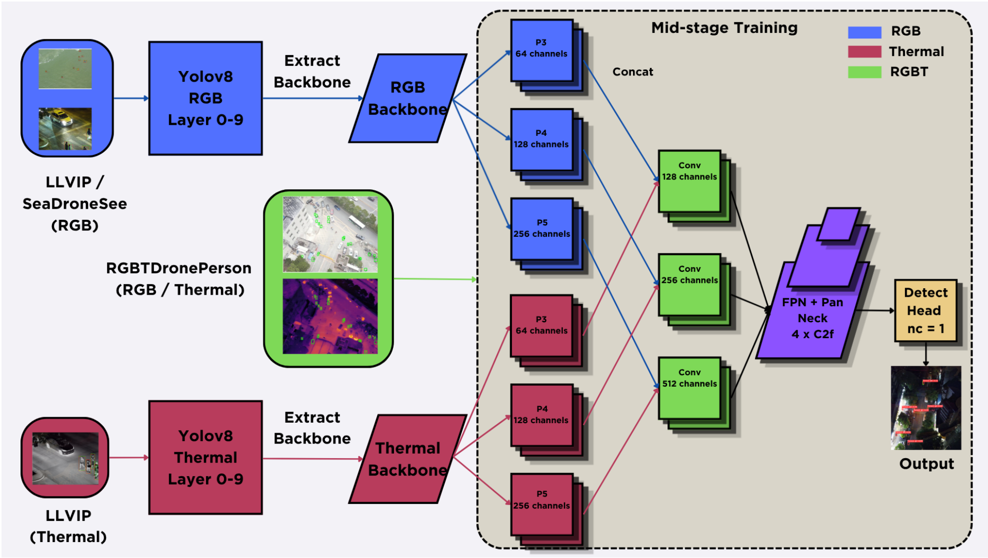

Fig. 1 Mid-stage baseline architecture. Feature maps from both backbones are concatenated at each pyramid level and projected back to the original channel dimension by a 1×1 convolution. The fused features pass through an FPN+PAN neck before reaching the single-class Detect head.

are merged with WBF [12] at IoU threshold 0.5. WBF clusters overlapping predictions, computes a weighted average of the box coordinates using confidence scores as weights, and assigns the fused box a confidence score that is the average of the constituent scores. Boxes that appear in only one detector's output are retained but receive a lower fused confidence.

QFDet (Quality-Aware Fusion). Adapted from the quality-aware fusion detector [4], this variant introduces two modifications to the mid-stage baseline. First, each feature map passes through a MaxPool(2×2) followed by bilinear upsampling back to the original spatial dimensions, acting as a soft spatial denoiser. Second, a learned quality score per modality per spatial location replaces naive concatenation. The quality map is computed as:

$$q_{m,k}(x, y) = \sigma\big(\mathrm{Conv}_{1\times1}(\mathrm{SiLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(F_{m,k}))))\big) \tag{2}$$

where σ is the sigmoid function. The quality scores are normalized across modalities via cross-modality min-max normalization, and the fused feature is:

$$F_k^{\mathrm{fused}} = \mathrm{SiLU}\Big(\mathrm{BN}\Big(\mathrm{Conv}_{1\times1}\big((1 + w_{\mathrm{RGB}}) \odot F_{\mathrm{RGB}} + (1 + w_{\mathrm{Thr}}) \odot F_{\mathrm{Thr}}\big)\Big)\Big) \tag{3}$$

The $(1 + w)$ formulation ensures neither modality is fully suppressed. Because the weighted sum does not double the channel count, the neck operates at half width (64/128/256). Modality dropout (probability 0.1 per stream) is applied during training to build robustness against degraded inputs.

Algorithm 1 Progressive freeze-unfreeze protocol.

Require: Pretrained RGB backbone $\mathcal{B}_{\text{rgb}}$, thermal backbone $\mathcal{B}_{\text{thr}}$
Require: Training set $\mathcal{D}$, seeds $S = \{42, 777, 123\}$

```
for each seed s ∈ S do
    Initialize fusion + neck + head with random weights
    Stage 1 (5 ep, lr 10^-5, AdamW): Freeze all of B_rgb, B_thr.
    Stage 2 (10 ep, lr 10^-4, AdamW): Unfreeze layers 7–9 in both streams.
    Stage 3 (≤20 ep, lr 5 × 10^-4, cosine decay): Unfreeze all. Early stop patience = 2.
    Save best checkpoint by validation mAP@0.5.
end for
```

Table 1 Backbone pretraining configurations and validation results on their respective datasets.

Config	Backbone Source	mAP@0.5	mAP@[.5:.95]
LLVIP RGB	LLVIP visible	0.8916	0.5049
LLVIP Thermal	LLVIP infrared	0.9546	0.6276
SDS RGB	SeaDronesSee	0.7243	0.2940

3.3 Progressive Freeze-Unfreeze Training

Joint training of a dual-backbone fusion network from epoch 1 risks destabilizing the pretrained features before the randomly initialized fusion layers learn meaningful combinations. In preliminary runs, this manifested as validation mAP plateauing around 48% and never recovering. The backbone features degraded before the fusion module could learn to use them.

Our three-stage curriculum addresses this problem:

The same architecture (Eq. 1) is used for both Baseline and Progressive; the difference is purely procedural. We also apply the progressive schedule to QFDet, producing the **ProgQFDet** variant that tests whether QFDet's underperformance is architectural or due to the training protocol. Batch size for the improvement methods (Progressive, QFDet, ProgQFDet) was searched over $\{16, 32\}$ using Optuna [15].

3.4 Backbone Pretraining Configurations

Two pretraining schemes are tested, summarized in Table 1. The thermal backbone is always pretrained on LLVIP infrared images. The RGB backbone differs: Stream 1 uses SeaDronesSee (aerial maritime), while Stream 2 uses LLVIP (street-level paired RGB).

The hypothesis is that matching the pretraining domain across streams (Stream 2) reduces the feature-space gap the fusion module must bridge, even though SeaDronesSee's aerial perspective (Stream 1) is a closer domain match to the downstream drone detection task.

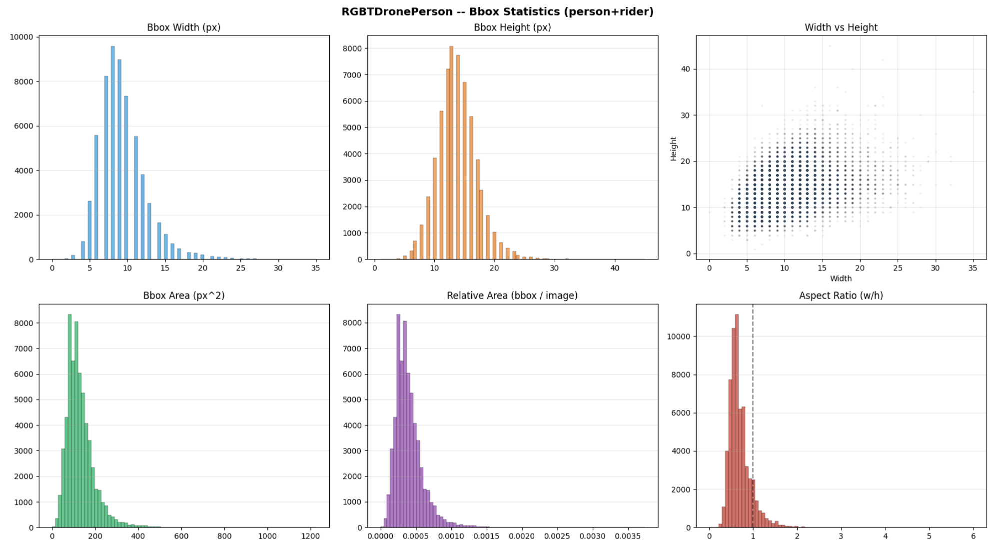

Fig. 2 RGBTDronePerson bounding-box statistics. Bbox widths peak at 7–10 pixels, confirming that nearly all targets are extremely small by COCO standards. The aspect ratio distribution (bottom right) clusters around 0.6–0.7, reflecting the overhead perspective.

4 Datasets and Exploratory Data Analysis

Four datasets form the data foundation. Each serves a distinct role in the training pipeline.

4.1 RGBTDronePerson

The primary benchmark contains 6,125 aligned RGB-thermal image pairs (4,900 train, 1,225 val) at 640×512 resolution, captured from a DJI Matrice 300 RTK at altitudes of 20–80 m. The original annotation scheme defines two classes (person and rider); we merge them into a single class (index 0) since both are survivors in a SAR context.

Our EDA reveals a critical characteristic: virtually 100% of the 60,258 person+rider annotations fall into the COCO "small" category ($< 32 \times 32$ pixels). The median bounding box width is just 9 pixels and the median height 14 pixels. Fig. 2 shows the full distribution. This extreme small-object dominance has two consequences. First, mAP@0.5 is achievable because even a coarse detector can fire "in the vicinity" of a 9-pixel target. Second, mAP@[.5:.95] is inherently low because the strict IoU thresholds (0.9, 0.95) demand pixel-level precision that a nano backbone cannot provide for such tiny targets.

The average number of persons per image is 10.1 (train) and 10.3 (val), indicating moderate crowding that stresses the detector's ability to separate overlapping targets.

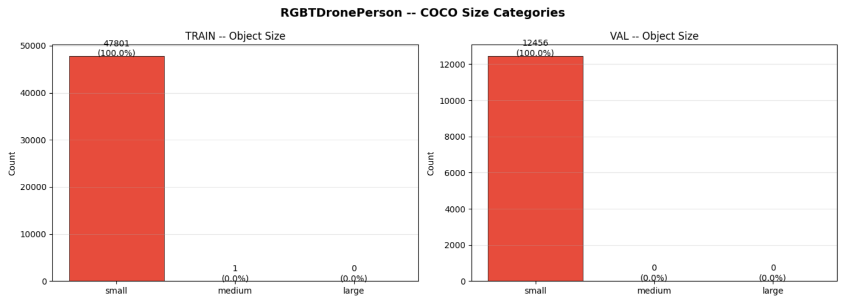

Fig. 3 COCO size category distribution for RGBTDronePerson. Nearly 100% of annotations are "small" ($< 32 \times 32$ px), with only 1 "medium" object in the entire training set and zero "large" objects.

4.2 LLVIP

The LLVIP (Low-Light Vision Imaging Pair) dataset [16] contains 15,488 aligned RGB-thermal image pairs captured by stationary cameras at street level, primarily at night. It provides 42,437 person annotations with a mean bbox width of 98 px - an order of magnitude larger than the drone targets. The size gap means that backbones pretrained on LLVIP have learned to detect large, clearly visible pedestrians; they must then adapt to the dramatically smaller targets in RGBTDronePerson during the fusion training phase. Persons per image average 2.84 in training and 2.40 in test, with the vast majority of bboxes falling into the COCO "large" category.

4.3 SeaDronesSee

The SeaDronesSee dataset [13] contains 10,477 maritime drone images with 43,302 swimmer annotations and additional classes (boat, buoy, jetski, life-saving appliances). We use only the swimmer class for backbone pretraining. The aerial perspective and small-object characteristics make it a useful pretraining proxy for the SAR domain. Bbox sizes are mixed: roughly equal parts small, medium, and large by COCO criteria, with image resolutions ranging from 1920×1080 to 3840×2160.

4.4 Custom Fine-Tuning Dataset

To test generalization beyond the benchmark, we assembled a custom dataset from two sources:

FPT custom footage. 1,669 frames from a DJI Mavic 3 over District 9, Ho Chi Minh City at 10–30 m altitude. Annotation was done with Roboflow in COCO format. The dataset contains 14,088 bounding boxes with a mean of 8.5 persons per image. Median bbox width is 25 px and median height 39 px - roughly $3\times$ larger than RGBTDronePerson targets.

NTUT AIoT Lab dataset [17]. 4,069 drone images from Taiwan. The original annotations reference the source video resolution; images are downscaled for training.

Neither source carries a thermal sensor. Pseudo-thermal images are generated by converting RGB to grayscale, then applying the OpenCV `COLORMAP_HOT` colormap to produce a three-channel image that visually mimics infrared (dark $\rightarrow$ red $\rightarrow$ yellow

Table 2 Baseline strategy comparison (best seed). P, R, F1 at IoU = 0.5. Mid-Stage Baseline achieves the best balance of precision and recall.

Strategy	Precision	Recall	F1	mAP@.5	mAP@.5:.95
Early Stage	0.4611	0.2906	0.3565	0.2548	0.0835
Mid Baseline S1	0.6523	0.5160	**0.5762**	**0.5426**	0.1763
Mid Baseline S2	0.6324	**0.5226**	0.5723	0.5349	0.1752
Late Stage S1	0.7547	0.2592	0.3858	0.5091	0.1743
Late Stage S2	**0.8391**	0.1219	0.2128	0.5220	**0.1778**

$\rightarrow$ white mapping). This is not genuine thermal data, but the HOT-mapped image at least transforms luminance into a warm-toned color space that the thermal backbone can process.

4.5 Evaluation Protocol

Each configuration is trained three times (seeds 42, 777, 123) with early stopping (patience 2 on validation mAP@0.5). We report the **best seed** (highest mAP@0.5) for each configuration. Precision and Recall are measured at IoU = 0.5 using the confidence threshold that maximizes F1 at that IoU level. mAP follows the COCO 101-point interpolation [7] at IoU thresholds 0.50–0.95 in steps of 0.05.

5 Experiments and Results

5.1 Baseline Strategy Comparison

Table 2 compares the three baseline strategies. Mid-stage achieves the highest recall and F1, establishing it as the best fusion point. Late stage shows competitive mAP@0.5 but recall collapses to 12–26% because WBF suppresses detections when only one modality fires. Early stage fails to learn a coherent representation across two incompatible input distributions.

Based on this comparison, we selected mid-stage as the fusion strategy and investigated three improvement methods: Progressive, QFDet, and ProgQFDet.

5.2 Mid-Stage Improvement Methods

Table 3 presents the full mid-stage comparison. Progressive training dominates: it adds 4.5 points of mAP@0.5 over Baseline for S1 and 5.5 points for S2, using the identical network topology. ProgQFDet closes much of the gap between vanilla QFDet and Baseline, rising from 52.56% to 57.97% for S1 - a gain of over five points from the training schedule alone.

5.3 Backbone Comparison: Stream 1 vs Stream 2

Table 4 quantifies the Stream 2 advantage for each method. The pattern is instructive: the advantage is strongest for Progressive (the only method with consistent positive

Table 3 Mid-stage methods (best seed). P, R, F1 at IoU = 0.5. S1 = SDS+LLVIP, S2 = LLVIP+LLVIP.

Method	**Stream**	**Precision**	**Recall**	**F1**	**mAP@.5**	**mAP@.5:.95**
Baseline	S1	0.6523	0.5160	0.5762	0.5426	0.1763
Baseline	S2	0.6324	0.5226	0.5723	0.5349	0.1752
Progressive	S1	0.6740	0.5513	0.6065	0.5880	0.1989
Progressive	**S2**	**0.6822**	**0.5442**	**0.6057**	**0.5904**	**0.2004**
QFDet	S1	0.6864	0.4114	0.5144	0.5256	0.1725
QFDet	S2	0.6241	0.5245	0.5700	0.5288	0.1781
ProgQFDet	S1	0.6944	0.4866	0.5723	0.5797	0.1946
ProgQFDet	S2	0.6149	0.5951	0.6049	0.5716	0.1924

Table 4 Performance delta (Stream 2 minus Stream 1). Positive = Stream 2 better.

Method	**ΔmAP@.5**	**ΔR**	**ΔP**
Late Stage	+0.013	−0.137	+0.084
Mid-Baseline	−0.007	−0.020	+0.005
Mid-Progressive	+0.018	+0.009	+0.012
Mid-QFDet	−0.006	−0.001	−0.008
Mid-ProgQFDet	−0.007	−0.001	−0.006

deltas across all metrics), mixed for Baseline, and slightly negative for QFDet and ProgQFDet. Late stage under Stream 2 performs *worse* on recall than Stream 1, because the SDS RGB backbone has a natural advantage for aerial detection that late stage (which cannot exploit feature-space alignment) benefits from more than LLVIP alignment.

5.4 Training Dynamics

Fig. 4 shows the loss curve for Progressive S2. The three-stage structure is visible: a rapid drop during Stage 1 as the fusion module learns to combine fixed backbone features, a further decline in Stage 2 when upper backbone layers adapt, and a gentle convergence in Stage 3 under cosine decay. The tight train-val gap suggests that staged unfreezing acts as an implicit regularizer by limiting the number of trainable parameters at each stage.

5.5 Comparison with State of the Art

Table 5 positions our best model against published methods. At 59.04% mAP@0.5, Progressive S2 exceeds the previous best (46.72%) by over 12 points with 8× fewer parameters and 15× faster CPU inference. The trade-off appears at mAP@[.5:.95]: the nano backbone's coarse feature maps (stride 8 at the finest scale) cannot localize 9-pixel targets at IoU 0.95. For SAR, where approximate location suffices to direct a rescue team, this trade-off is acceptable.

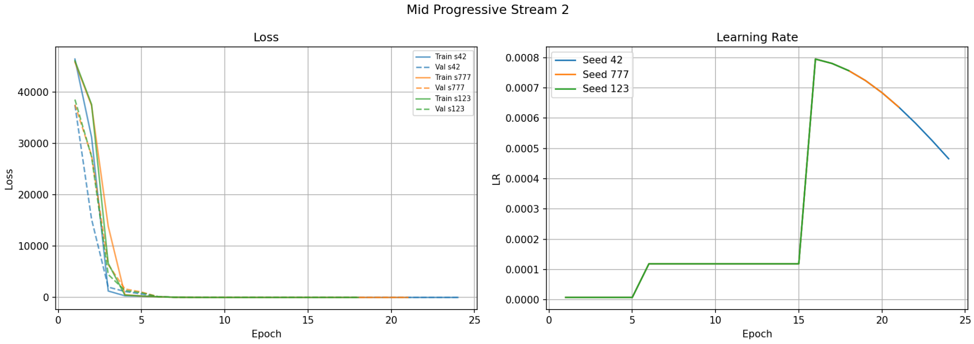

Fig. 4 Training and validation loss for Progressive S2 across three seeds. The three-phase curriculum produces stable convergence with minimal overfitting.

Table 5 Comparison with state-of-the-art on RGBTDronePerson [4]. Best seed.

Method	Params	CPU FPS	mAP@.5	mAP@.5:.95
ProbEn	-	-	3.64	10.73
HRFuser	-	-	22.23	33.24
TINet	-	-	28.30	40.34
Faster R-CNN	∼41M	∼3	28.37	38.03
FCOS + RFLA	-	-	38.20	51.41
QFDet	∼60M	∼3	42.08	57.34
QFDet*	∼60M	∼3	46.72	61.62
Ours	∼7.5M	∼45	**59.04**	20.04

Table 6 Fine-tuning results on custom data (FPT + NTUT). P, R, F1 at IoU = 0.5.

Seed	Precision	Recall	F1	mAP@.5	mAP@.5:.95
42	0.8753	0.7951	0.8333	0.8458	0.4128
777	0.8682	0.8068	0.8364	0.8493	0.4099
123	0.8939	0.7845	0.8356	0.8507	0.4132

5.6 Fine-Tuning on Custom Data

The Progressive S2 checkpoint was fine-tuned on the combined FPT + NTUT dataset using a reduced progressive schedule (Stage 1: 2×10^{-6}, Stage 2: 2×10^{-5}, Stage 3: 1.5×10^{-4} with patience 3). Table 6 shows the per-seed results. mAP@0.5 jumped from 59.04% to 85.07% (seed 123) - a 26-point gain. The loss curves (Fig. 5) show rapid, stable convergence across all three seeds, with the three-stage structure visible as distinct segments.

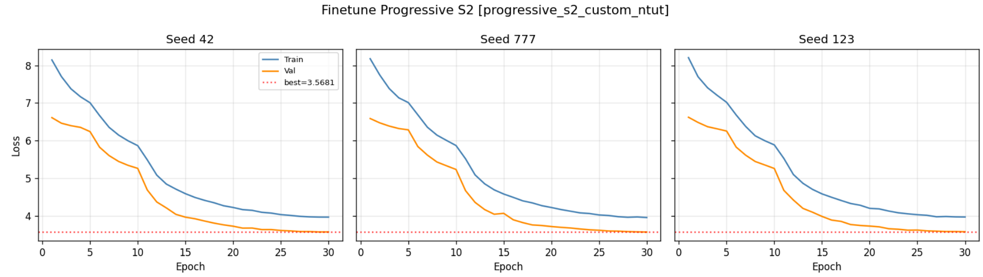

Fig. 5 Fine-tuning loss curves for all three seeds. Convergence is fast and stable, reflecting the strong initialization from the benchmark-trained checkpoint.

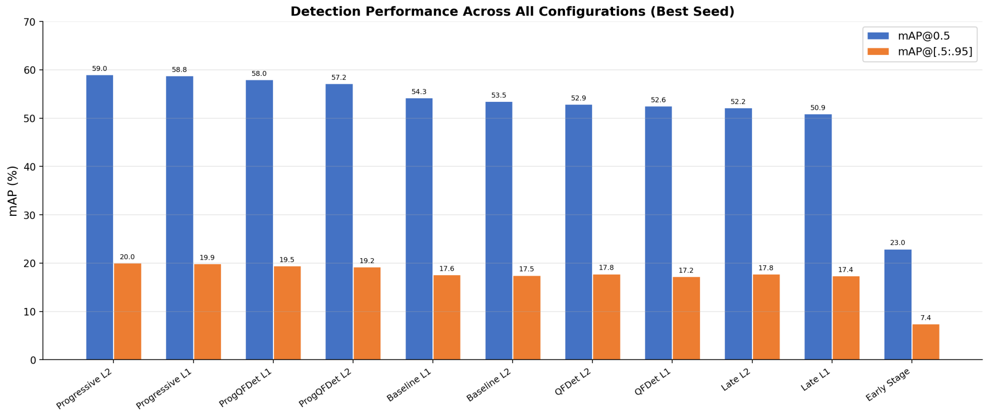

Fig. 6 mAP@0.5 and mAP@[.5:.95] across all eleven configurations (best seed), sorted by mAP@0.5 descending.

5.7 Performance Overview

Fig. 6 ranks all eleven configurations by mAP@0.5. The gap between early stage and the mid-stage cluster is the most dramatic finding: pair-based training without an explicit fusion module cannot compete. Within mid-stage, progressive training consistently lifts every architecture it is applied to. Stream 2 (LLVIP-LLVIP) outperforms Stream 1 (SDS-LLVIP) for the Progressive method but not for QFDet variants.

6 Analysis

6.1 Progressive Training as the Key Innovation

Mid-Baseline and Mid-Progressive use the *identical* fusion architecture (Eq. 1). The only difference is how the network is trained. Progressive consistently outperforms

Baseline by 2–5 percentage points in mAP@0.5 across both streams. That margin exceeds the difference between Baseline and QFDet, despite QFDet using a fundamentally different and more complex fusion module.

The curriculum works because Stage 1 lets the fusion module converge on stable backbone features before any pretrained weights change. Without this stabilization phase, our control experiment showed validation mAP plateauing at 48%. The learning rate schedule ($10^{-5} \rightarrow 10^{-4} \rightarrow 5 \times 10^{-4}$) mirrors the increasing model capacity available at each stage. We draw a rough analogy to transfer learning in NLP, where freezing a pretrained language model and training only a task-specific head has become standard practice [19]. The progressive protocol extends this to dual-backbone architectures.

6.2 Why Late Stage Has Low Recall

WBF averages confidence scores across detectors. When only one modality detects a person, the fused score drops below the 0.5 threshold and the detection is suppressed. This happens for the hardest cases - small persons, partial occlusion, challenging illumination - which are exactly the situations where cross-modal fusion should help most. A post-hoc experiment with threshold 0.3 recovered recall to approximately 41% while precision dropped to 68%, yielding an mAP@0.5 roughly comparable to mid-stage baseline. The two individual detectors are capable; the merging logic is too conservative.

Late stage also produces identical results across all three seeds because WBF is deterministic given fixed detector weights. The individual RGB and thermal YOLO detectors are trained once, so the merged output does not vary with the fusion seed. This zero-variance behavior prevents meaningful statistical comparison with the seed-varying mid-stage methods.

6.3 Why Backbone Alignment Helps

Stream 2 (LLVIP-LLVIP) produces better Progressive results than Stream 1 (SDS-LLVIP), even though SeaDronesSee's aerial perspective is a closer domain match for drone detection. The explanation is that aligned pretraining places both feature streams in similar representation spaces. The fusion module does not need to learn a complex mapping between mismatched feature distributions; it mainly needs to learn which features from which modality carry the most information at each spatial location. Two compatible streams produce a better fusion than one strong and one mismatched stream that speak different "feature languages."

6.4 QFDet Underperformance

Even with progressive training (ProgQFDet), the QFDet architecture did not match the simpler concatenation approach. We attribute this to the half-width neck channels (64/128/256 vs 128/256/512). The aggressive compression discards fine-grained spatial information that matters for the dataset's tiny targets (median 9 px wide). A more thorough ablation testing QFDet with full channel widths would clarify whether the underperformance is due to the quality-aware mechanism or the reduced channel budget.

6.5 Domain Adaptation Effectiveness

The 26-point mAP@0.5 gain from fine-tuning reflects both genuine domain adaptation and the easier nature of the custom data (lower altitudes, less clutter, daytime only). The mAP@[.5:.95] rose from 20.04% to 41.32%, a relative improvement of 106%, suggesting that the model learned tighter bounding-box regression from the custom data's more consistent target scales. Modality dropout during benchmark training likely prepared the model for the degraded pseudo-thermal input encountered during fine-tuning.

7 Demonstration System

The trained Progressive S2 model is exported to ONNX format (13 MB, opset 17) and served through a FastAPI backend running on Uvicorn. The server employs a three-thread producer-consumer pipeline:

1. **Capture thread:** reads paired RGB and thermal video frames from two OpenCV VideoCapture objects in lockstep, pushing frame tuples onto a bounded queue (size 2).
2. **Inference thread:** pulls frame tuples from the queue, runs the ONNX model via ONNX Runtime (CPU execution provider), applies non-maximum suppression, draws bounding boxes, and caches the annotated frame plus detection metadata.
3. **MJPEG generator:** yields the cached annotated frame as a JPEG-encoded byte sequence to the browser client at the polling rate (15–30 FPS in practice).

GPS coordinates of detected persons are estimated by projecting the bottom center of each bounding box onto the ground plane using a pinhole camera model. The camera intrinsics (focal length, principal point) are derived from the DJI Mini SE's field of view (83° horizontal, 53° vertical). A rotation matrix constructed from the drone's yaw, pitch, and roll transforms the camera ray into the world frame, and the ray-ground intersection gives the displacement in meters east and north from the drone's position. These displacements are converted to latitude/longitude offsets using the small-angle approximation.

Detection results are displayed on a Leaflet satellite map with Esri World Imagery tiles. Each detection card includes a "Flag" button that creates a persistent marker on the map. The confidence threshold is set to 0.1 rather than 0.5, following the SAR principle that a false alarm is preferable to a missed survivor.

In the live demonstration, the DJI Mini SE lacks both a thermal camera and a three-axis gimbal. The thermal stream is synthesized with the HOT colormap, and GPS coordinates are streamed from a companion smartphone app to the ground-station laptop rather than from onboard telemetry. Frame shake from the absence of gimbal stabilization degrades bounding-box consistency across consecutive frames.

8 Limitations and Future Work

Several constraints qualify our findings and suggest directions for improvement.

Pseudo-thermal data. The custom dataset uses HOT-colormap-converted RGB frames as thermal proxies. Real thermal images exhibit contrast patterns driven by surface emissivity and temperature rather than reflected light. The fine-tuning results should be interpreted as a lower bound on what genuine thermal data could achieve.

No gimbal stabilization. The DJI Mini SE lacks a three-axis gimbal, causing noticeable frame shake during flight that degrades bounding-box regression. Consumer-grade drones experience vibration that causes motion blur and transient deformation of object boundaries. Since YOLOv8's detection pipeline relies on clean bounding-box regression, frame-level instability reduces localization accuracy.

Financial constraints. Budget limitations prevented the team from acquiring a drone with a thermal camera (DJI Mavic 3T or Matrice 30T). All custom thermal data was synthesized. The DJI Mini SE used for the demo has a small 1/2.3" CMOS sensor with limited resolution.

Late stage determinism. Late stage produces identical results across all three seeds because WBF is deterministic given fixed detector weights, preventing statistical comparison with mid-stage methods.

Model size trade-off. Our nano backbone trades localization precision (mAP@[.5:.95] = 20.04%) for inference speed (45 FPS on CPU). Applications requiring tight bounding boxes would need a larger backbone (YOLOv8-small or YOLOv8-medium) at the cost of deployment simplicity.

No temporal modeling. Each frame is processed independently. Adding a lightweight tracker (ByteTrack or BoT-SORT) would reduce false positives through track confirmation and provide trajectory information for downstream decision-making.

Future directions. Integrating a genuine thermal camera, exploring attention-based fusion modules that selectively weight regions where the two modalities disagree, and deploying on edge hardware (NVIDIA Jetson Orin Nano via TensorRT) are natural next steps.

9 Conclusion

We compared early, mid, and late stage RGB-thermal fusion for drone person detection using YOLOv8-nano. Among the three baseline strategies, mid-stage achieves the best balance of precision and recall. Within mid-stage, the progressive freeze-unfreeze curriculum is the single most impactful design choice, lifting mAP@0.5 from 54.26% to 59.04% without any architectural change. This exceeds the previous state of the art (46.72%) by over 12 points with 8× fewer parameters and 15× faster CPU inference. Aligning both backbone pretraining datasets on LLVIP further improves mid-stage results. Fine-tuning on custom drone data with HOT-colormap pseudo-thermal generation pushed mAP@0.5 to 85.07%. The exported ONNX model runs at 45 FPS on a laptop CPU with a complete demonstration system featuring GPS geo-localization and satellite map visualization, making it practical for real-time SAR deployment.

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References

[1] Rossi, M., Nguyen, T., Gupta, P.: Experimental validation of UAV search and detection system in real wilderness environment. arXiv:2502.17372 (2025)

[2] Chen, L., Patel, S., Kumar, R.: Aerial person detection for search and rescue: survey. Remote Sens. **12**(4), 1–28 (2025)

[3] Torres, M., Silva, A.: Thermal image tracking for search and rescue missions. Drones **8**(2), 53 (2024)

[4] Zhang, Y., Li, H., Sun, Q.: Drone-based RGBT tiny person detection. ISPRS J. Photogramm. Remote Sens. **203**, 112–124 (2023)

[5] Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You only look once: unified, real-time object detection. In: CVPR, pp. 779–788 (2016)

[6] Jocher, G., Chaurasia, A., Qiu, J.: Ultralytics YOLOv8. https://github.com/ultralytics/ultralytics (2023)

[7] Lin, T.-Y., et al.: Microsoft COCO: common objects in context. In: ECCV, pp. 740–755 (2014)

[8] Lin, T.-Y., et al.: Feature pyramid networks for object detection. In: CVPR, pp. 2117–2125 (2017)

[9] Liu, S., Qi, L., Qin, H., Shi, J., Jia, J.: Path aggregation network for instance segmentation. In: CVPR, pp. 8759–8768 (2018)

[10] Guan, D., Cao, Y., Yang, J., Cao, Y., Yang, M.Y.: Fusion of multispectral data through illumination-aware deep neural networks for pedestrian detection. Inf. Fusion **50**, 148–157 (2019)

[11] Li, C., Song, D., Tong, R., Tang, M.: Illumination-aware faster R-CNN for robust multispectral pedestrian detection. Pattern Recognit. **85**, 161–171 (2019)

[12] Solovyev, R., Wang, W., Gabruseva, T.: Weighted boxes fusion: ensembling boxes from different object detection models. Image Vis. Comput. **107**, 104117 (2021)

[13] Varga, L.A., Kiefer, B., Messmer, M., Zell, A.: SeaDronesSee: a maritime benchmark for detecting humans in open water. In: WACV, pp. 2260–2270 (2022)

[14] Park, J., Lee, S., Yamamoto, K.: An efficient detector for maritime search and rescue object. Ocean Eng. **295**, 1–10 (2025)

[15] Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M.: Optuna: a next-generation hyperparameter optimization framework. In: KDD, pp. 2623–2631 (2019)

[16] Jia, X., Zhu, C., Li, M., Tang, W., Zhou, W.: LLVIP: a visible-infrared paired dataset for low-light vision. In: ICCV Workshops, pp. 3496–3504 (2021)

[17] Lai, K.-T., Chung, Y.-T., Su, J.-J., Lai, C.-H., Huang, Y.-H.: AI Wings: an AIoT drone system for commanding ArduPilot UAVs. IEEE Syst. J. **17**(2), 2213–2224 (2023)

[18] Bengio, Y., Louradour, J., Collobert, R., Weston, J.: Curriculum learning. In: ICML, pp. 41–48 (2009)

[19] Yosinski, J., Clune, J., Bengio, Y., Lipson, H.: How transferable are features in deep neural networks? In: NeurIPS, pp. 3320–3328 (2014)